

CEHv13 MODULE 8 – SNIFFING

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Introduction

Sniffing is the process of monitoring and capturing network traffic to analyse data packets transmitted over a network.

Attackers use sniffing to intercept sensitive information such as usernames, passwords, credit card details, and confidential communications.

Network Sniffing Fundamentals

What is Network Sniffing?

Network Sniffing is the practice of capturing and analyzing network traffic to extract sensitive information such as usernames, passwords, confidential data, and communication patterns. With sniffing, you can monitor all sorts of traffic, either protected or unprotected, by enabling promiscuous mode on the network interface.

Definition

Network Sniffing involves monitoring data packets as they traverse network segments by placing network interfaces in promiscuous mode. The attacker can reveal information such as usernames and passwords from captured traffic. Anyone within the same LAN can potentially sniff packets.

Types Of Sniffing

Passive Sniffing

- Capturing data without altering the network traffic.
- Usually done on a hub or in wireless mode.
- The attacker listens silently to network traffic that is broadcast to all devices on the network segment.
- Difficult to detect because no packets are injected or the traffic altered.

Active Sniffing

- Actively injecting traffic to redirect packets to the attacker.
- Used in switched network environments where traffic is directed only to the intended port.

- Techniques include ARP spoofing, MAC flooding, and DHCP attacks to manipulate switch behavior.
- Allows interception of traffic not normally sent to the attacker's machine but is more detectable due to network interference.

How Network Sniffers Work

Working Mechanism

1. **Network Connection:** The attacker connects to the target network to start sniffing operations
2. **Promiscuous Mode:** Sniffers turn the Network Interface Card (NIC) into promiscuous mode
3. **Packet Capture:** In promiscuous mode, the NIC responds to every packet it receives, not just those destined for its MAC address
4. **Data Extraction:** The attacker decrypts and analyzes packets to extract valuable information

Network Infrastructure Analysis

Switch vs Hub Behaviour

Switch Characteristics

1. **Unicast Handling:** Forwards unicast packets only to specific destination ports
2. **Broadcast/Multicast:** Forwards broadcast and multicast traffic to all ports
3. **CAM Table:** Maintains MAC address table for intelligent forwarding
4. **Collision Domain:** Each port represents a separate collision domain

Hub Characteristics

1. **Shared Medium:** All devices share the same collision domain
2. **Broadcast Nature:** Transmits all packets to all connected ports
3. **Security Risk:** Makes sniffing easier due to shared medium
4. **Legacy Technology:** Largely replaced by switches in modern networks

Switch Port Analyser (SPAN) Port

SPAN Port (also known as **Port Mirroring**) is a switch feature that sends a copy of network packets from one or more switch ports to a designated monitoring port for analysis.

Uses:

- Network troubleshooting and analysis
- Security monitoring and intrusion detection
- Traffic analysis and performance monitoring
- Compliance and forensic investigations

Advanced Sniffing Concepts

Wiretapping

Wiretapping involves gaining information by intercepting signals from communication lines such as telephone circuits or internet connections. This activity is typically performed by third parties for surveillance purposes.

Types of Wiretapping

Active Wiretapping

Monitoring and Recording: Captures communication data

Signal Alteration: Modifies the communication during interception

Real-time Manipulation: Can inject or modify data streams

Higher Detection Risk: More likely to be discovered due to signal changes

Passive Wiretapping

1. **Silent Monitoring:** Records information without altering communication
2. **No Signal Changes:** Maintains original communication integrity
3. **Stealth Operation:** Harder to detect due to no modifications
4. **Data Collection:** Focuses on gathering intelligence without interference

Lawful Interception/Wiretapping

1. **Legal Authorisation:** Conducted with proper legal warrants
2. **Law Enforcement:** Performed by authorised security agencies
3. **Compliance Requirements:** Must follow strict legal procedures
4. **Court Orders:** Requires judicial approval and oversight

Common Sniffing Techniques

- **ARP Poisoning** – Manipulating ARP tables to intercept packets by sending forged ARP replies, redirecting network traffic to the attacker.
- **MAC Flooding** – Overloading the switch MAC address table with fake addresses, causing the switch to enter a fail-open hub mode, which broadcasts traffic to all ports.
- **DNS Spoofing** – Redirecting DNS queries to malicious servers, leading victims to fake websites or malicious content.
- **MITM (Man-in-the-Middle)** – Intercepting and possibly altering the communication between two parties without their knowledge.
- **DHCP Spoofing** – Providing malicious IP configuration information to clients, redirecting traffic or disrupting network access.

Tools for Sniffing:

- **Wireshark** – Network packet analysis tool with real-time capture, deep inspection, and extensive protocol support.
- **Tcpdump** – Command-line packet capture and analysis for Unix-like systems, highly customizable with powerful filtering options.
- **Ettcap** – Tool for man-in-the-middle attacks and network sniffing, supporting active and passive modes.
- **dsniff** – Suite of tools to sniff passwords, capture credentials, and conduct man-in-the-middle attacks.
- **Cain & Abel** – Password recovery tool on Windows with sniffing capabilities to intercept network credentials.

Network Sniffing Tools

Wireshark

Comprehensive network protocol analyser that captures and displays packet data in real-time, supporting hundreds of protocols with powerful filtering and analysis capabilities.

Example Usage:

Start packet capture on a specific interface

Wireshark -i eth0

Capture to file for later analysis

wireshark -i eth0 -w network_capture.pcap

Command-line capture with tshark

tshark -i eth0 -f "tcp port 80" -w http_traffic.pcap

Analyse captured file with filters

tshark -r network_capture.pcap -Y "http.request.method == POST"

Extract HTTP objects from capture

tshark -r network_capture.pcap --export-objects http,extracted_files/

Test against the target site

tshark -i eth0 -f "host rnivqlhaedmb4yc3ezjzgj7xy3prff4.oastify.com"

TCPDUMP

Command-line packet analyser that captures network traffic on Unix-like systems, providing powerful filtering options and output formats.

Example Usage:

Basic packet capture

tcpdump -i eth0

Capture HTTP traffic

tcpdump -i eth0 'tcp port 80'

Capture and save to file

tcpdump -i eth0 -w capture.pcap

Read from saved file

tcpdump -r capture.pcap

Capture with ASCII output

tcpdump -i eth0 -A 'tcp port 80'

Capture specific host traffic

tcpdump -i eth0 host 192.168.1.100

Monitor target site traffic

tcpdump -i eth0 'host rnivqlhaedmb4yc3ezjzgj7xy3prff4.oastify.com'

...

Ettercap

Comprehensive suite for man-in-the-middle attacks on LAN networks, supporting ARP poisoning, DNS spoofing, and traffic interception.

Example Usage:

ARP poisoning attack

```
ettercap -T -M arp:remote /192.168.1.1// /192.168.1.100//
```

DNS spoofing with target redirection

```
echo "*.target-site.com A 192.168.1.50" >> /etc/ettercap/etter.dns
```

```
echo "rnivqlhaedmb4yc3ezjzgj7xy3prff4.oastify.com A 192.168.1.50" >> /etc/ettercap/etter.dns
```

```
ettercap -T -M arp:remote -P dns_spoof /192.168.1.1// /192.168.1.100//
```

SSL stripping attack

```
ettercap -T -M arp:remote -P sslstrip /192.168.1.1// //
```

GUI mode for easier operation

```
ettercap -G
```

NetworkMiner

Network forensics tool that extracts artifacts from network traffic captures, including files, images, messages, and credentials.

Example Usage:

Start NetworkMiner GUI application

```
mono NetworkMiner.exe
```

Command-line version for automated analysis

```
mono NetworkMinerCLI.exe -r capture.pcap -o output_directory
```

Extract files from PCAP

```
mono NetworkMinerCLI.exe --pcap capture.pcap --output extracted_files/
```

Analyse target site traffic

```
mono NetworkMinerCLI.exe -r target_traffic.pcap --keyword "oastify.com"
```

Bettercap

Modern network attack and monitoring framework with support for WiFi, Bluetooth, and network reconnaissance.

Example Usage:

Start interactive session
sudo bettercap

ARP spoofing configuration
arp.spoof on
set arp.spoof.targets 192.168.1.100

→DNS spoofing with target redirection
set dns.spoof.domains *.target-site.com,*.oastify.com
set dns.spoof.address 192.168.1.50
dns.spoof on

HTTP proxy with script injection
set http.proxy.script inject.js
http.proxy on

WiFi reconnaissance
wifi.recon on

Capture specific traffic
set net.sniff.filter "host rnivqlhaedmb4yc3ezjzgj7xy3prff4.oastify.com"
net.sniff on

Dsniff Suite

Collection of tools for network auditing and penetration testing, including dsniff, urlsnarf, filesnarf, and mailsnarf.

Example Usage:

Password sniffing from network traffic
dsniff -i eth0

URL monitoring and logging
urlsnarf -i eth0

File extraction from network streams
filesnarf -i eth0

Email monitoring and capture
mailsnarf -i eth0

ARP spoofing with arpspoof
arpspoof -i eth0 -t 192.168.1.100 192.168.1.1

Monitor target site connections
`dsniff -i eth0 -s 1024 | grep "oastify.com"`

Countermeasures and Detection for Sniffing Attacks:

Network Security Best Practices

Switch Security Configuration

- Port Security: Limit MAC addresses per port to prevent unauthorized devices.
- DHCP Snooping: Disable rogue DHCP servers by validating DHCP messages.
- ARP Inspection: Verify ARP packets to block spoofing attacks.
- VLAN Segmentation: Isolate network segments to reduce attack surface.

Monitoring and Detection

- Deploy Network Monitoring Tools such as IDS/IPS systems to detect suspicious traffic patterns.
- Establish Baselines of normal network traffic for anomaly detection.
- Use Anomaly Detection to identify unusual network behaviors indicating sniffing.
- Regularly review and analyze network logs for indications of attacks.

Encryption and Authentication

- Implement End-to-End Encryption (e.g., SSL/TLS, VPN) to protect data in transit.
- Perform Certificate Validation to avoid man-in-the-middle attacks.
- Enforce Strong Authentication mechanisms like multi-factor authentication (MFA).
- Use VPNs to secure remote network connections.

Additional Measures

- Regularly update and patch network devices and endpoints.
- Educate users on risks associated with public Wi-Fi and phishing.
- Detect devices in promiscuous mode and block unauthorised sniffing tools.
- Conduct periodic vulnerability assessments and penetration testing to identify sniffing exposures.

Tools for Sniffing

TOOLS FOR SNIFFING --

Active Sniffing --

MAC FLOODING--

1. MACOF (Kali)
2. Scapy (kali)
3. Hping3 (Kali)
4. Ettercap (Kali)
5. Bettercap (kali)

DHCP Starvation --

1. Yersinia (kali)
2. DHCPig (kali)
3. DHCPStrav (kali)

ARP Poisoning

1. Cain and able (windows)
2. XARP Application --Detection ARP Poisoning (Windows)
3. Ettercap (kali)
4. Bettercap (Kali)
5. ARP Poisoning

MAC Spoofing

1. TMAcV6 (Windows)
2. Macchanger (kali)
3. SMAC (Windows)
4. Mac Address Changer (Windows)

Passive Sniffing -- only capture and monitoring packets

1. Wireshark
2. TCP Dump